

Task Redistribution System

A case study in designing fairer household work around burden, changing capacity, and trustworthy automation

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PROJECT STATUS: This is a product concept, not a completed research study. No formative interviews, diary study, usability study, algorithm calibration, or longitudinal pilot has been completed. Research and testing are work to do next, and the case study should be updated as evidence is collected.

Abstract

Most household task tools are built to remember, assign, and track chores. This concept asks a different question: what if the same task is easy for one person, draining for another, and harder for either person on a low-capacity day? The Task Redistribution System is a proposed household planning tool that treats fairness as the distribution of burden rather than the equal division of task counts. It considers visible work, planning and monitoring, personal friction, current capacity, deadlines, recent workload, and household agreements before suggesting who should take a task. The first version deliberately uses simple, visible decision rules instead of opaque artificial intelligence. Users can see why a suggestion was made, correct the system, snooze or transfer work within limits, and retain control over what personal information is shared. The project is designed as a broader exploration of trustworthy decision systems: how software can represent subjective human experience without pretending it is perfectly measurable, and how automation can support negotiation without becoming surveillance, judgment, or control.

Keywords: household labor, fairness, mental load, capacity, explainability, privacy, trustworthy automation, product strategy

1. The Product Problem

A chore list can tell a household that two people each completed five tasks. It cannot tell whether those five tasks required the same amount of effort, planning, discomfort, interruption, or responsibility. That difference matters because household work is not only the visible action of doing something. Someone must often notice that a task exists, remember when it is due, decide how it should be done, gather what is needed, perform it, and check that it stays handled. Research on cognitive household labor describes this planning and monitoring work as a meaningful part of domestic responsibility [1].

The product therefore starts from a simple thesis: equal task counts are not the same as equal burden. A fair system should be able to recognize that people experience tasks differently and that a fair arrangement can legitimately change from one week to the next.

CORE THESIS: Fair does not necessarily mean 50/50. The goal is not to prove that one person worked harder. The goal is to make the work visible enough that two people can negotiate and redistribute it with less guesswork.

What the product is - and is not

- It is a shared planning and recommendation system for household responsibilities.
- It is not a scorecard for deciding who is the better partner, roommate, or household member.
- It can recognize personal friction - such as sensory discomfort, activation difficulty, emotional strain, or strong dislike - without requiring a diagnosis or medical label.
- It does not need to decide what is objectively fair. Instead, it helps users make their assumptions visible, agree on a fairness approach, and revise it when circumstances change.

2. Design Thesis: Model Burden, Not Just Tasks

The central design challenge is to represent enough of a person's experience to make useful recommendations without turning ordinary life into a complicated scoring exercise. The system therefore uses a small set of plain-language signals. They are not treated as scientific measurements. They are user-provided estimates that help the system compare options and explain its reasoning.

Signal	What it means in everyday language
Visible effort	How much time or physical work the task usually takes.
Planning and ownership	How much remembering, deciding, scheduling, checking, or follow-up is required.
Personal friction	How unpleasant or difficult this task feels for this particular person. This can include sensory discomfort, task initiation difficulty, emotional strain, or strong dislike.

Signal	What it means in everyday language
Current capacity	How much a person can reasonably take on right now. A demanding week, illness, exams, travel, caregiving, or poor sleep may temporarily lower capacity.
Timing and constraints	Whether the task has a hard deadline, requires a specific location or account, or can only be completed by certain people.
Recent load	Whether one person has already carried more of the household burden recently, including repeated high-friction or planning-heavy work.

This model is intentionally broader than time. Time is easy to count, but a ten-minute task can still create more friction than a thirty-minute one. The system needs to acknowledge that difference without claiming it can measure human effort perfectly.

Why current capacity is separate from long-term fairness

A household can have a long-term goal of roughly equal contribution while still allowing temporary imbalance. If one person is sick, in an exam week, or dealing with a demanding work period, the system should be able to protect short-term capacity without rewriting the household's long-term agreement. This prevents a bad week from being treated as a permanent preference and prevents long-term fairness from becoming daily debt collection.

3. How the Redistribution Logic Works

In this case study, “algorithm” simply means the set of rules the product follows when it makes a suggestion. The first version should be easy enough that a user can understand the recommendation without knowing computer science.

1. First, the system looks at what needs to be done and which tasks are urgent, recurring, or dependent on another task.
2. It removes options that are not possible. For example, a task may require a car, access to a particular account, physical presence, or a skill that only one person currently has.
3. For every remaining person, it considers how burdensome that task is likely to be for them, using the household task information and that person's private friction settings.
4. It then looks at current capacity. A person who has temporarily marked themselves as low-capacity should receive less optional work, especially work that is flexible or high-friction.
5. The system checks recent workload so that the easiest assignment in the moment does not repeatedly push the same person into becoming the default owner of difficult work.
6. It recommends the option that appears to create the most reasonable balance, then gives a short explanation in ordinary language.
7. The recommendation remains a suggestion. Either person can accept it, change it, correct the task information, or choose another arrangement.

EXAMPLE EXPLANATION: “Suggested for Alex this week because the task is flexible for Alex, Sam marked low capacity today, and Sam handled most of the planning-heavy work over the last several days.” The interface should show only the factors necessary to explain the decision - not private notes or sensitive reasons.

Why the first version should be rule-based

A rule-based system is easier to inspect, test, and challenge. That matters because the product is making recommendations about a relationship, not sorting files in the background. Before the product learns from behavior, the team needs to know whether users understand the rules, whether the rules capture the right kinds of burden, and whether people feel safe disagreeing with the system. A more adaptive model is only valuable if it improves the experience enough to justify the added complexity and privacy risk.

4. Core Product Experience

Onboarding: establish a household agreement before automating

The system should not begin by asking dozens of personal questions. The minimum useful setup is: who is participating, what recurring responsibilities exist, whether some tasks have fixed owners or restrictions, and what the household broadly means by fair. Each person can then privately mark a small number of tasks that are unusually easy or unusually difficult for them. More detailed calibration should be optional and introduced only when it clearly improves recommendations.

Invisible labor becomes visible work

Planning, remembering, and monitoring should not disappear behind the person who performs the final visible action. The product can represent “owning groceries,” for example, as more than physically shopping. Ownership may include noticing staples are low, maintaining a list, coordinating preferences, deciding when to go, and making sure the household is stocked. This prevents the app from rewarding only the visible final step while hiding the responsibility that surrounds it.

Snooze and transfer without turning flexibility into avoidance

People need to be able to say “not now” without breaking the system. A snooze should therefore be bounded by the task's real deadline rather than treated as failure. A transfer should require the receiving person to accept it; tasks should never silently move to someone else. Repeated transfers can be surfaced during a weekly review so the household can decide whether the original ownership, task definition, or burden assumptions are wrong.

Weekly review instead of constant negotiation

The daily product should stay lightweight: see what matters, understand the recommendation, complete or adjust it. More sensitive questions belong in a weekly review. That review can ask whether the week felt fair, which tasks caused friction, whether capacity assumptions were accurate, and whether any responsibilities should be redefined. This creates a place for negotiation without requiring users to debate every recommendation in the moment.

Optional incentives should be treated cautiously

An earlier concept included an optional “savings penalty,” where repeated opting out could move a user-chosen amount of money into a savings goal. This should not be part of the first product. Money can quickly turn a flexible household tool into punishment, and it may work very differently across relationships and income levels. If the idea is ever explored, it should be opt-in, reversible, capped, and researched specifically for coercion, resentment, and unintended pressure.

5. Fairness Without Pretending Fairness Is Objective

There is no single correct definition of household fairness. Some households may want a roughly equal long-term burden. Others may prefer an arrangement based on available time, alternating ownership, personal strengths, or an explicitly unequal split during certain life stages. The product should not hide these choices inside its code. It should ask the household to choose or discuss the rule it wants to follow.

This is important because fairness in a social system cannot be reduced to one number [2]. A system can be mathematically consistent and still feel unfair if it ignores invisible work, treats one person's preferences as less legitimate, or turns a temporary situation into permanent responsibility. The product therefore treats its recommendation as an aid to negotiation, not a final verdict.

DESIGN RULE: The app should never say “this is fair.” It should say what it considered, what it recommends, and what the household can change.

Avoiding the expertise trap

A common household pattern is that the person who is faster or more experienced at a task keeps receiving it. That can make the system efficient while locking one person into permanent ownership. The recommender should therefore recognize repeated assignment and occasionally suggest learning, rotation, or shared ownership when the task is safe to rotate. Efficiency is one consideration, not the only one.

6. Privacy, Safety, and Abuse-Aware Design

The information that makes this product useful can also make it sensitive. A person's capacity, disliked tasks, private reasons, patterns of avoidance, or emotional friction can reveal more than they intend. The product should therefore be private by default and collect only what it needs. Privacy is not only about preventing an outside data breach; it is also about preventing unnecessary exposure between people who share the same household [3,4].

Design choice	Why it matters
Private personal settings	A partner can know that a task is currently high-friction without seeing the private reason behind that rating.
Minimal explanation	Explain enough to understand a recommendation, but do not reveal sensitive notes or build a dossier about a person.
No hidden monitoring	Do not infer effort from location, device activity, calendars, or sensors unless a user deliberately enables a clearly explained integration.
Easy override and correction	Users must be able to disagree with a recommendation and correct the information that produced it.
No public scoreboards	The product should not rank household members, assign “good partner” scores, or create evidence that can easily be weaponized in conflict.
Data separation and exit	If a household changes or a relationship ends, each person should be able to leave, export their own information, and remove shared access cleanly.

The misuse question is especially important. A fairness tool can become harmful if one person uses it to pressure another, demand private explanations, or treat system suggestions as proof that someone is lazy. Abuse-aware design means the product should never

pretend its estimate is objective evidence about a person. It should be designed to support consent and negotiation even when the users do not have equal power.

7. Important Edge Cases

A good product concept should explain what happens when the simple version of the problem breaks. The following cases are important because they test whether the system supports real life rather than an idealized household.

Situation	Product response
Both people strongly dislike the same task	Do not force a fake “fair” answer. Suggest rotation, batching, outsourcing if feasible, changing the standard, or negotiating an explicit owner.
One person marks almost everything as high-friction	Treat the ratings as information, not wrongdoing. Ask whether the scale is being interpreted differently and offer recalibration rather than silently discounting that person.
One person is always faster	Avoid assigning the task to that person forever. Separate short-term efficiency from long-term ownership and learning.
People disagree on what “done” means	Create a shared definition of completion for that task before using performance or repetition data.
One person has low capacity for several weeks	Prompt a household-level conversation about the temporary target instead of accumulating invisible “debt.”
A task can only be done by one person	Treat it as a real constraint and balance elsewhere rather than pretending the task can be redistributed.
The relationship or household ends	Stop shared recommendations, separate accounts and data, and make exit simple and non-punitive.

8. Why the Concept Stands Out

The differentiation is not that no other household app tracks effort, rotates chores, or discusses mental load. Existing tools already address parts of those problems. The stronger product hypothesis is the combination of several ideas that are usually separated: person-specific burden, changing capacity, invisible ownership work, transparent redistribution, and privacy-aware explanation.

- It treats burden as personal. The same task does not have to carry the same cost for everyone.
- It treats capacity as temporary. The system can respond to a difficult week without permanently changing what “fair” means.
- It treats planning and monitoring as work, not as invisible support around the “real” task.
- It explains recommendations instead of presenting an unexplained assignment.
- It preserves human control. Suggestions can be corrected, overridden, snoozed, transferred, or renegotiated.
- It treats privacy and relationship misuse as product requirements rather than afterthoughts.
- It does not require AI to sound sophisticated. The first goal is a decision process people can understand and trust.

DIFFERENTIATION HYPOTHESIS: This combination may be meaningfully different from ordinary chore tracking, but that is still a hypothesis. Competitive research and user research should test whether people notice, value, and understand the difference.

9. Where AI Could Help - and Where It Should Not

This can be an AI product strategy case study even before the first version uses machine learning. The important product problem is deciding how much decision-making to automate, what information an automated system should be allowed to use, how it communicates uncertainty, and how users recover when it is wrong. Those are the same questions that matter in higher-stakes AI and security products.

Useful future applications

- Learn from explicit corrections, such as when users repeatedly say that a task is harder or easier than the default estimate.
- Notice recurring overload patterns and suggest a household review before the same conflict repeats.
- Suggest task bundles, rotations, or schedule changes that reduce repeated friction.
- Summarize weekly patterns in neutral language so users can discuss them without reading raw logs.
- Adapt explanations to show the few factors that actually mattered for a recommendation.

Boundaries for adaptive systems

The system should not infer diagnoses, relationship quality, motivation, or moral character from task behavior. It should not secretly learn personal preferences from sensitive signals and should not make large changes without showing users what changed. If a simple rules system performs just as well, keeping the simpler system may be the better product decision. This follows the broader principle that trustworthy AI should be valid for its intended use, understandable, privacy-aware, and controllable [3].

10. What I Would Build First

The minimum viable product should test whether the core idea is useful before building sophisticated automation. A first prototype only needs enough functionality to make burden visible and support a better household decision.

Build first	Defer until validated
Shared recurring task list and ownership	Automatic calendar, location, or device monitoring
Simple task information: effort, planning, deadline, flexibility	Complex personalized machine-learning models
Private person-specific friction settings	Diagnosis-based scoring or labels
Simple current-capacity control	Passive inference of mood, stress, or health
Transparent recommendations with plain-language reasons	Gamified leaderboards and partner scores
Accept, override, snooze, and transfer flows	Savings penalties or financial stakes
Weekly fairness and correction review	Automatic long-term behavior changes without confirmation

The success criterion for this stage is not “the algorithm was correct.” It is whether the tool helps people understand the work, discuss tradeoffs, and make changes without creating more administrative effort or conflict than it removes.

11. Research and Validation Roadmap

Because no product research has been completed yet, the current model should be treated as a set of design hypotheses. The next version of this case study should replace assumptions with evidence where possible and clearly document what changed.

Stage	What I would learn	Why it changes the product
1. Desk research and competitive audit	How existing products represent chores, ownership, effort, mental load, fairness, and household negotiation.	Prevents false claims of novelty and identifies where the concept actually needs to differentiate.
2. Formative interviews	How people currently divide work, what they consider unfair, which responsibilities stay invisible, and which information they would or would not share with a partner or app.	Tests whether the burden categories reflect real language and real problems.
3. Short diary study	What people notice, plan, do, postpone, hand off, and remember across normal days rather than only in retrospective interviews.	Shows whether the model captures changing capacity and invisible labor over time.
4. Prototype usability study	Whether people can set up tasks, understand recommendations, find privacy controls, correct mistakes, and use snooze/transfer without confusion.	Tests comprehension and interaction before investing in more automation.
5. Decision-logic calibration	Whether the initial rules create recommendations users consider reasonable across different household patterns.	Identifies which factors matter, which overlap, and which should be removed or simplified.
6. Longitudinal pilot	Whether the tool reduces repeated negotiation or simply creates a new layer of household administration.	Tests sustained value, fairness perceptions, misuse, abandonment, and whether adaptive features are justified.

Questions that should remain open until research

- Do users experience “dislike,” sensory friction, activation difficulty, and emotional friction as meaningfully different inputs, or would one simpler difficulty setting work better?
- Should planning and monitoring be represented as separate tasks, part of ownership, or both?
- Do people want the system to assign tasks, rank options, or simply show an imbalance and let them negotiate?
- How much history is useful before the experience starts to feel like debt accounting?
- What explanation is enough to create trust without exposing private information or overwhelming the task card?
- Do people actually want an adaptive system to learn their patterns, and if so, what should require explicit permission?

UPDATE RULE: Every future portfolio version should distinguish between external evidence, product assumptions, observed user behavior, and measured outcomes. If research contradicts the current concept, the design should change rather than forcing the evidence to fit the original idea.

12. What Success Would Mean

If the concept is tested, success should be measured through the household experience rather than task completion alone. A product that increases completion but also increases resentment, surveillance, or administrative work would not be a successful outcome.

- Users can explain why a recommendation was made and know how to change it.
- People report that previously invisible work is easier to discuss and redistribute.
- Temporary low-capacity periods can be accommodated without creating a sense of permanent debt.
- Users override recommendations when appropriate without feeling that they are “breaking” the system.
- The app does not become another responsibility disproportionately managed by one household member.
- Private information remains understandable and controllable, including what a partner can see.
- The system does not increase conflict, shame, coercion, or scorekeeping.
- Any future adaptive model performs meaningfully better than the simple rules version on user-centered outcomes, not just prediction accuracy.

13. Why This Project Fits My Broader Work

I want this project to show more than the ability to design a consumer productivity app. Its deeper value is that it lets me work on the same questions that interest me in psychological security, usable security, and AI product strategy: how people interpret system decisions, how sensitive context changes what “correct” means, how automation affects trust and behavior, and what happens when a system's assumptions do not match a person's lived circumstances.

Portfolio area	What this case study demonstrates
Psychological security	The design considers how system language, scoring, visibility, and automation can create pressure, shame, trust, resistance, or perceived control.
Usable security and privacy	Sensitive data is minimized, explanations reveal only what is necessary, users can correct the model, and misuse within a shared environment is treated as a design concern.
AI security / responsible AI product management	The project asks when automation is justified, which signals should be off-limits, how recommendations should expose uncertainty, and how users recover from wrong decisions.
Product strategy	A broad interpersonal problem is translated into a product thesis, explicit assumptions, staged MVP, future intelligence roadmap, research gates, and success criteria.
Research mindset	The document separates evidence from hypotheses and defines what must be studied before stronger product claims are made.

That makes the project relevant to the type of work I want to do professionally: designing systems where human behavior, security, trust, and intelligent automation meet. The household domain keeps the problem understandable, while the underlying design questions generalize to higher-stakes systems.

14. Conclusion

The Task Redistribution System is built around a deliberately modest idea: software can help people reason about household work, but it should not pretend to know the objective truth about effort or fairness. The product makes burden more visible, accounts for changing capacity, recommends rather than commands, and keeps sensitive context under user control. Its first version should be transparent and simple. Research should determine which parts of the model are genuinely useful, which are unnecessary, and whether adaptive intelligence eventually adds value.

The strongest outcome for this project is not a complicated algorithm. It is a trustworthy decision system that people can understand, challenge, and reshape as their circumstances change.

References

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Research integrity note: This document contains conceptual product design, externally motivated problem framing, and proposed research. It contains no completed user research, usability results, algorithm calibration results, or product outcome claims for this concept.